THE ROOT SPACES OF THE EXTENDED GRAPH FORM THE IRREDUCIBLE INTEGRABLE HIGHEST WEIGHT MODULE OF HIGHEST WEIGHT W

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Allowed standard prerequisites (editorial).

- The extended graph obtained by adding the vertex infinity to the Dynkin graph, its Cartan matrix, and the simple root alpha_infinity. Introduced in the paragraph preceding the counted statement (source lines 1178-1182). Fixes the ambient Lie algebra and the root spaces whose direct sum the statement identifies with an irreducible highest weight module.
- Integrability of a highest weight module and the standard facts about Kac-Moody root multiplicities. Used inside the counted proof at source lines 1213-1215, attributed by the author to the cited book of Kac, sections 3.5 and 10.4. Supplies the integrability of the constructed highest weight module and the identification of the root multiplicities with the weight multiplicities.

Source reference. Original section 2.2: sources/algebra-fl01-quiver-varieties-and-branching/pinned-original-source.tex, source lines 726–726. Appears only in the author's introductory sentence before the counted statement, which is outside the delivered spans; the stub is not used by the counted material.

Lemma 0.1. *Let α_∞ denote the simple root corresponding to the vertex ∞. Let V be the direct sum of root spaces $\tilde{\mathfrak{g}}_\alpha$ of $\tilde{\mathfrak{g}}$, where α is of the form $\alpha = -\sum_{i\in I} m_i\alpha_i - \alpha_\infty$. Let $\mathfrak{g}$ act on V by the restriction of the adjoint representation. Then V is isomorphic to the irreducible integrable highest weight representation $V(\mathbf{w})$ with the highest*

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weight vector f_∞ of weight $\mathbf{w}$. In particular, the root multiplicity $\dim \tilde{\mathfrak{g}}_\alpha$ is equal to the weight multiplicity of $\alpha|_{\mathfrak{h}}$ in $V(\mathbf{w})$, where $\mathfrak{h}$ is the Cartan subalgebra of $\mathfrak{g}$.

Proof. From the definition we have

$$[e_i, f_\infty] = 0, \quad [h_i, f_\infty] = w_i f_\infty \qquad \text{for } i \in I.$$

Moreover $\tilde{\mathfrak{g}}_\alpha$ is a linear span of the elements of the form

$$[f_{i_1}, [f_{i_2}, [\dots, [f_{i_{k-1}}, [f_\infty, \underbrace{[f_{i_k}, [\dots [f_{i_{s-1}}, f_{i_s}] \dots]]}_{=x}]]] \dots]]] = -[f_{i_1}, [f_{i_2}, [\dots, [f_{i_{k-1}}, [x, f_\infty]] \dots]]]$$

for $i_1, i_2, \dots \in I$. This means that V is a highest weight module. We also know V is integrable by [2, 3.5]. Now the assertion follows from [2, 10.4]. □

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